TSAR Project Assignment Part 3 - Group 3

Times Series Forecasting

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2026

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1 Preliminaries

```
# Install needed libraries to use this report
if (!require(dplyr)) install.packages("dplyr")
if (!require(tidyr)) install.packages("tidyr")
if (!require(ggplot2)) install.packages("ggplot2")
if (!require(fpp3)) install.packages("fpp3")

#Load needed libraries
library(dplyr)
library(ggplot2)
library(tidyr)
library(fpp3)


# Read already filtered time series (tsibble)
# with Series ID CEU1000000001 (Mining and Logging)
mySeries <- readRDS("us_employment_filtered.rds")
```

1.1 Overview of tsibble

```
# Display the first few rows of the tsibble
head(mySeries)
```

```
# A tsibble: 6 x 4 [1M]
# Key:       Series_ID [1]
     Month Series_ID     Title               Employed
     <mth> <chr>         <chr>                  <dbl>
1 1939 Jan CEU1000000001 Mining and Logging      885
2 1939 Feb CEU1000000001 Mining and Logging      882
3 1939 Mär CEU1000000001 Mining and Logging      883
4 1939 Apr CEU1000000001 Mining and Logging      614
5 1939 Mai CEU1000000001 Mining and Logging      719
6 1939 Jun CEU1000000001 Mining and Logging      852
```

```
# Inspect the structure of the tsibble (index, keys, column data types)
str(mySeries)
```

```
tbl_ts [969 x 4] (S3: tbl_ts/tbl_df/tbl/data.frame)
 $ Month    : mth [1:969] 1939 Jan, 1939 Feb, 1939 Mär, 1939 Apr, 1939 Mai, 1939 Jun...
 $ Series_ID: chr [1:969] "CEU1000000001" "CEU1000000001" "CEU1000000001" "CEU1000000001"
```

```
 $ Title    : chr [1:969] "Mining and Logging" "Mining and Logging" "Mining and Logging" "
 $ Employed : num [1:969] 885 882 883 614 719 852 847 864 894 941 ...
 - attr(*, "key")= tibble [1 x 2] (S3: tbl_df/tbl/data.frame)
  ..$ Series_ID: chr "CEU1000000001"
  ..$ .rows    : list<int> [1:1]
  .. ..$ : int [1:969] 1 2 3 4 5 6 7 8 9 10 ...
  .. ..@ ptype: int(0)
  ..- attr(*, ".drop")= logi TRUE
 - attr(*, "index")= chr "Month"
  ..- attr(*, "ordered")= logi TRUE
 - attr(*, "index2")= chr "Month"
 - attr(*, "interval")= interval [1:1] 1M
  ..@ .regular: logi TRUE
```

```r
# Get a summary of the columns (distribution, range, missing values)
summary(mySeries)
```

```
     Month              Series_ID             Title              Employed
 Min.   :1939 Jan   Length:969         Length:969         Min.   : 556.0
 1st Qu.:1959 Mär   Class :character   Class :character   1st Qu.: 676.0
 Median :1979 Mai   Mode  :character   Mode  :character   Median : 765.0
 Mean   :1979 Mai                                         Mean   : 793.8
 3rd Qu.:1999 Jul                                         3rd Qu.: 899.0
 Max.   :2019 Sep                                         Max.   :1260.0
```

2 Model 1: Mean Method

2.1 Step 1: Extracting the Training Set

```r
# Extracting number of rows for training set with 80/20 split
overall_count <- (nrow(mySeries))
training_row_count <- floor((nrow(mySeries) * 0.8))
train <- mySeries %>% slice(1:training_row_count)
```

2.2 Step 2: Train 1st Benchmark Model

```r
# Mean method used
mean_employment_fit <- train %>%
  model(
  Mean = MEAN(Employed)
```

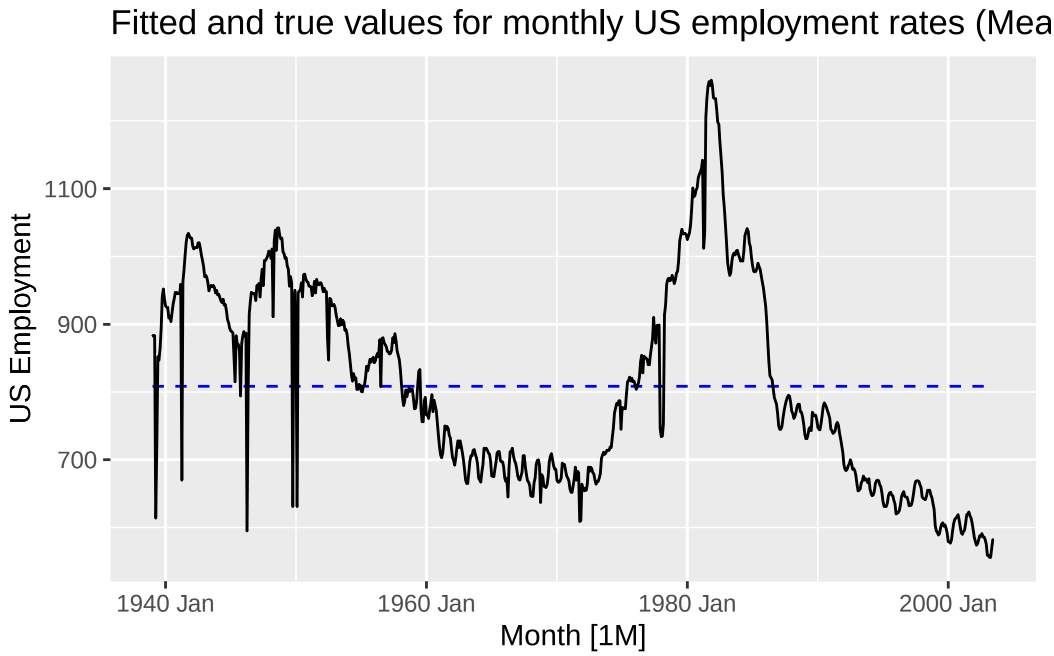
```
)

# View first impression of the fitted values
augment(mean_employment_fit)
```

```
# A tsibble: 775 x 7 [1M]
# Key:       Series_ID, .model [1]
   Series_ID     .model     Month Employed .fitted .resid .innov
   <chr>         <chr>      <mth>    <dbl>   <dbl>  <dbl>  <dbl>
 1 CEU1000000001 Mean    1939 Jan      885    809.   76.4   76.4
 2 CEU1000000001 Mean    1939 Feb      882    809.   73.4   73.4
 3 CEU1000000001 Mean    1939 Mär      883    809.   74.4   74.4
 4 CEU1000000001 Mean    1939 Apr      614    809. -195.  -195.
 5 CEU1000000001 Mean    1939 Mai      719    809.  -89.6  -89.6
 6 CEU1000000001 Mean    1939 Jun      852    809.   43.4   43.4
 7 CEU1000000001 Mean    1939 Jul      847    809.   38.4   38.4
 8 CEU1000000001 Mean    1939 Aug      864    809.   55.4   55.4
 9 CEU1000000001 Mean    1939 Sep      894    809.   85.4   85.4
10 CEU1000000001 Mean    1939 Okt      941    809.  132.   132.
# i 765 more rows
```

2.3 Step 3: Evaluate the Model Fit of 1st Benchmark Model

2.3.1 Task A

```
# Plot fitted values (predictions on training set) against actual values over time
augment(mean_employment_fit) %>%
  autoplot(.fitted, colour="blue", linetype="dashed") +
  autolayer(train, Employed) +
  labs(y= "US Employment", title="Fitted and true values for monthly US employment rates (
```



2.3.1.1 Answering questions to 1st Benchmark Model

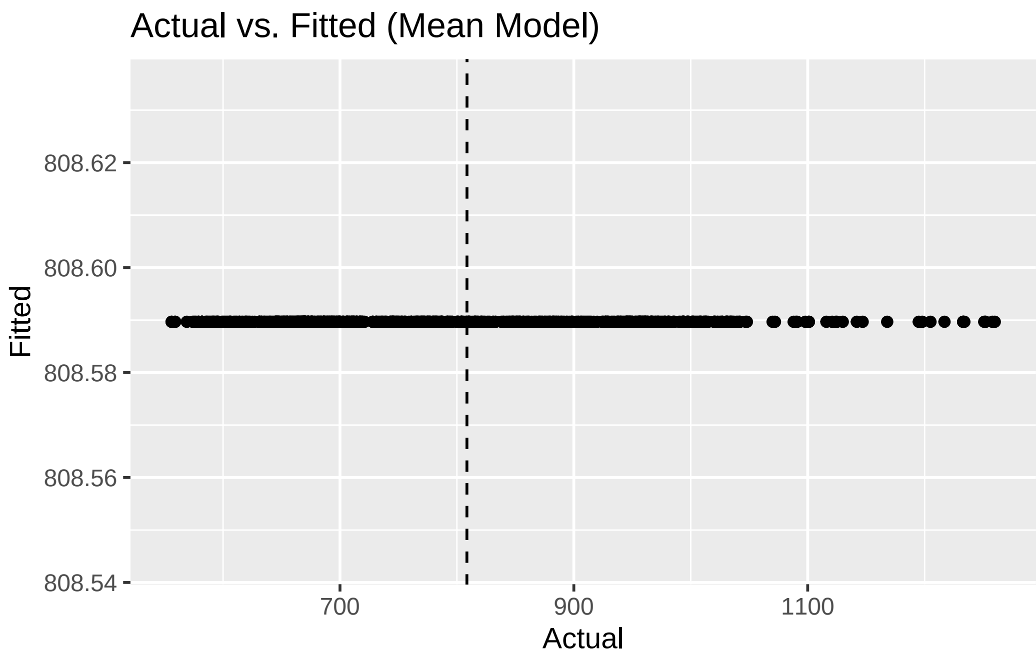
Are the fitted values close to the actual values?

Are they systematically too high or too low?

If yes, where does it come from? (Consider how your benchmark model works!)

2.3.2 Task B

```
# Create a scatter plot of actual values vs. fitted values with an identity line (slope=1)
augment(mean_employment_fit) %>%
  ggplot(aes(x=Employed, y=.fitted)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype=2) +
  labs(title = "Actual vs. Fitted (Mean Model)", x="Actual", y="Fitted")
```



2.3.2.1 Answering questions to 1st Benchmark Model

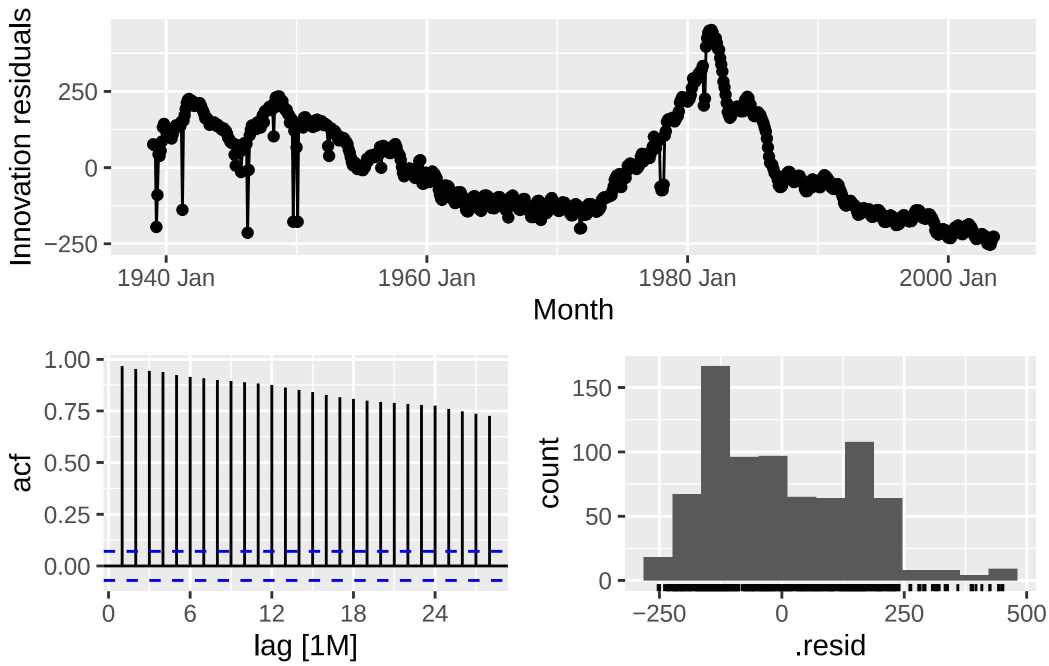
Are the points close to the identity line?

Are they systematically too high or too low?

If yes, where does it come from? (Relate your answer to what you saw in the time plot.)

2.3.3 Task C

```
# Generate diagnostic plots for the model residuals (time plot, ACF plot, histogram)
mean_employment_fit %>% gg_tsresiduals()
```



2.3.3.1 Answering questions to 1st Benchmark Model

Are the residuals auto-correlated? How do you decide that based on the plots?

Do the residuals have zero mean? How do you decide that based on the plots?

What do these results tell you about your model?

2.3.4 Task D

```
# Perform a Ljung-Box test for residual autocorrelation with 24 lags (2 * seasonal period)
augment(mean_employment_fit) %>%
  features(.innov, ljung_box, lag=24)
```

```
# A tibble: 1 x 4
  Series_ID      .model lb_stat lb_pvalue
  <chr>          <chr>    <dbl>     <dbl>
1 CEU1000000001 Mean     14183.         0
```

2.3.4.1 Answering questions to 1st Benchmark Model

Does the test result support your conclusions from (c)?

How do you conclude that from the test result?

```r
# Calculate the mean of the residuals of the fitted model
residual_mean <- mean(
  augment(mean_employment_fit)$.resid,
  na.rm = TRUE)
residual_mean
```

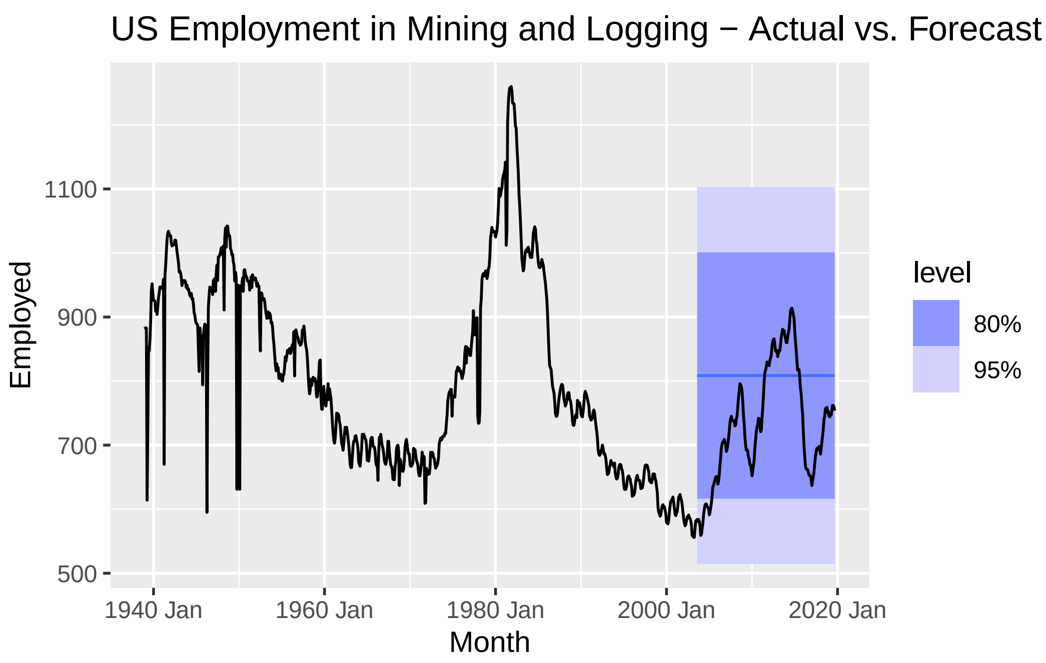
```
[1] -3.226362e-14
```

2.3.4.2 Answering questions to 1st Benchmark Model

Does the result support your conclusions from (c)?

2.4 Step 4: Evaluate the Point Forecast Accuracy of your 1st Benchmark Model

```r
# Make a forecast and save the result
fc_mean <- mean_employment_fit %>%
  forecast(h = (nrow(mySeries) - nrow(train)))

# Create a time plot 'Actual vs. Forecast'
fc_mean %>%
  autoplot(mySeries) +
  labs(title="US Employment in Mining and Logging - Actual vs. Forecast (Mean Model)")
```



```r
# Calculate the forecast accuracy metrics
fc_mean %>%
  accuracy(mySeries)
```

```
# A tibble: 1 x 11
  .model Series_ID     .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
  <chr>  <chr>         <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 Mean   CEU1000000001 Test  -74.0  116.  98.8 -11.7  14.6  2.27  1.70 0.987
```

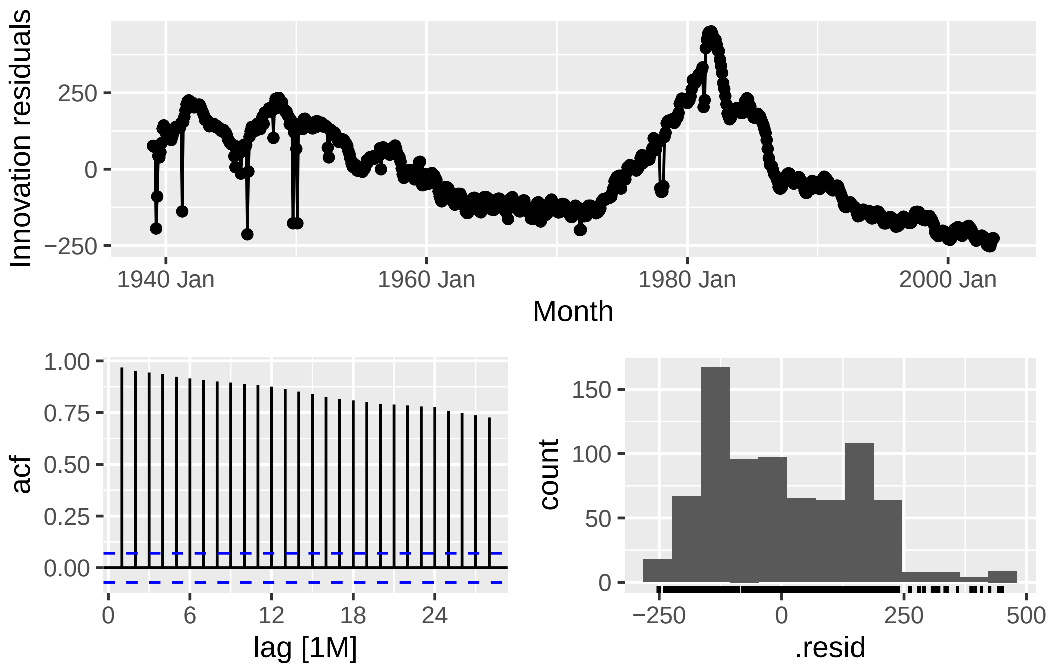
2.4.0.1 Answering questions to 1st Benchmark Model

Interpret the **RMSE** (Root Mean Squared Error) in max. 3 sentences:

Interpret the **MAPE** (Mean Absolute Percentage Error) in max. 3 sentences:

2.5 Step 5: Evaluate the Point Forecast Uncertainty of Your 1st Benchmark Model

```r
# Residual diagnostics for checking homoscedasticity and normality
mean_employment_fit %>% gg_tsresiduals()
```




```r
# Display prediction interval boundaries
fc_mean %>%
  hilo()
```

```
# A tsibble: 194 x 7 [1M]
# Key:       Series_ID, .model [1]
   Series_ID     .model    Month
   <chr>         <chr>     <mth>
 1 CEU1000000001 Mean   2003 Aug
 2 CEU1000000001 Mean   2003 Sep
 3 CEU1000000001 Mean   2003 Okt
 4 CEU1000000001 Mean   2003 Nov
 5 CEU1000000001 Mean   2003 Dez
 6 CEU1000000001 Mean   2004 Jan
 7 CEU1000000001 Mean   2004 Feb
 8 CEU1000000001 Mean   2004 Mär
 9 CEU1000000001 Mean   2004 Apr
10 CEU1000000001 Mean   2004 Mai
# i 184 more rows
# i 4 more variables: Employed <dist>, .mean <dbl>, `80%` <hilo>, `95%` <hilo>
```

2.5.0.1 Answering questions to 1st Benchmark Model

Are the residuals homoscedastic? (That is, do they have constant variance?)

Are the residuals normally distributed?

Does it make sense to include these prediction intervals in your model evaluation? Why/why not?

3 Model 2: Seasonal naive method with drift

3.1 Step 1: Extracting the Training Set

```
# Extracting number of rows for training set with 80/20 split
overall_count <- (nrow(mySeries))
training_row_count <- floor((nrow(mySeries) * 0.8))
train <- mySeries %>% slice(1:training_row_count)
```

3.2 Step 2: Train 2nd Benchmark Model

```
# Train the model with the seasonal naive drift method
naive_drift_employment_fit <- train %>% model (
    SNaive_Drift = SNAIVE(Employed ~ drift())
)
```

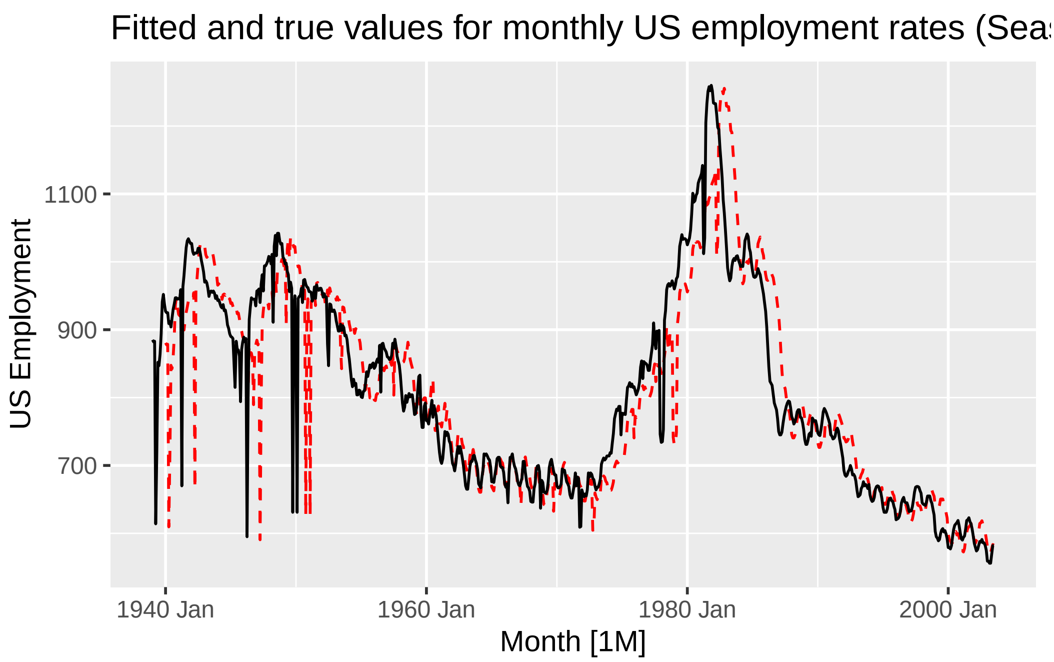
```r
# View the trained data
augment(naive_drift_employment_fit)
```

```
# A tsibble: 775 x 7 [1M]
# Key:       Series_ID, .model [1]
   Series_ID     .model           Month Employed .fitted .resid .innov
   <chr>         <chr>            <mth>    <dbl>   <dbl>  <dbl>  <dbl>
 1 CEU1000000001 SNaive_Drift 1939 Jan      885      NA     NA     NA
 2 CEU1000000001 SNaive_Drift 1939 Feb      882      NA     NA     NA
 3 CEU1000000001 SNaive_Drift 1939 Mär      883      NA     NA     NA
 4 CEU1000000001 SNaive_Drift 1939 Apr      614      NA     NA     NA
 5 CEU1000000001 SNaive_Drift 1939 Mai      719      NA     NA     NA
 6 CEU1000000001 SNaive_Drift 1939 Jun      852      NA     NA     NA
 7 CEU1000000001 SNaive_Drift 1939 Jul      847      NA     NA     NA
 8 CEU1000000001 SNaive_Drift 1939 Aug      864      NA     NA     NA
 9 CEU1000000001 SNaive_Drift 1939 Sep      894      NA     NA     NA
10 CEU1000000001 SNaive_Drift 1939 Okt      941      NA     NA     NA
# i 765 more rows
```

3.3 Step 3: Evaluate the Model Fit of Your 2nd Benchmark Model

3.3.1 Task A

```r
# Plot actual training data against the model's fitted values over time
augment(naive_drift_employment_fit) %>%
  autoplot(.fitted, colour="red", linetype="dashed") +
  autolayer(train, Employed) +
  labs(y= "US Employment", title="Fitted and true values for monthly US employment rates (
```



3.3.1.1 Answering questions to 2nd Benchmark Model

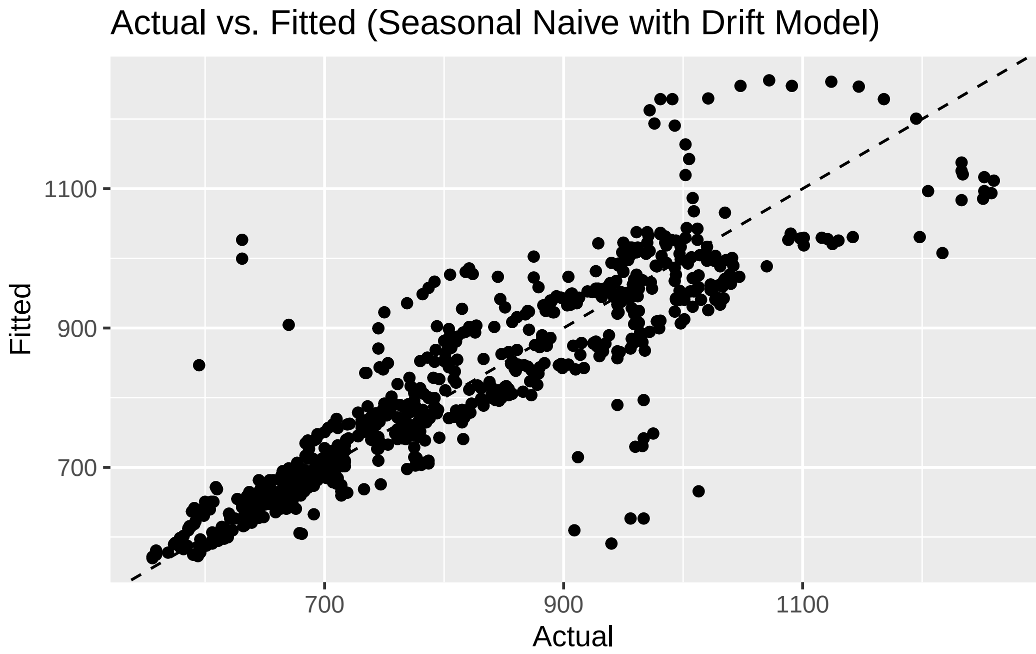
Are the fitted values close to the actual values?

Are they systematically too high or too low?

If yes, where does it come from? (Consider how your benchmark model works!)

3.3.2 Task B

```
# Plot actual vs. fitted values to check deviation from the identity line (slope=1)
augment(naive_drift_employment_fit) %>%
  ggplot(aes(x=Employed, y=.fitted)) +
  geom_point() +
  geom_abline(slope = 1, intercept = 0, linetype=2) +
  labs(title = "Actual vs. Fitted (Seasonal Naive with Drift Model)", x="Actual", y="Fitte
```



3.3.2.1 Answering questions to 2nd Benchmark Model

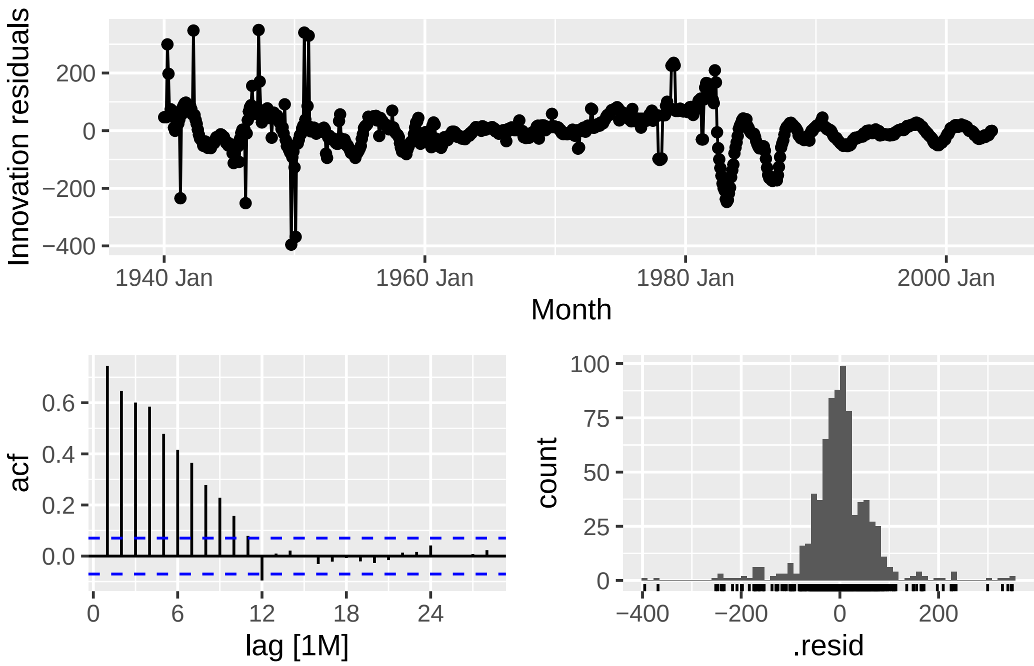
Are the points close to the identity line?

Are they systematically too high or too low?

If yes, where does it come from? (Relate your answer to what you saw in the time plot.)

3.3.3 Task C

```
# View residual diagnostic plots (time plot, ACF plot, histogram) to assess model fit
naive_drift_employment_fit %>% gg_tsresiduals()
```



3.3.3.1 Answering questions to 2nd Benchmark Model

Are the residuals auto-correlated? How do you decide that based on the plots?

Do the residuals have zero mean? How do you decide that based on the plots?

What do these results tell you about your model?

3.3.4 Task D

```
# Conduct a formal Ljung-Box test for autocorrelation in the residuals (lag = 2 * seasonal
augment(naive_drift_employment_fit) %>%
  features(
  .innov,
  ljung_box,
  lag = 24
)
```

```
# A tibble: 1 x 4
  Series_ID     .model        lb_stat lb_pvalue
  <chr>         <chr>           <dbl>     <dbl>
1 CEU1000000001 SNaive_Drift    1834.         0
```

3.3.4.1 Answering questions to 2nd Benchmark Model

Does the test result support your conclusions from (c)?

How do you conclude that from the test result?

```
# Calculate the mean of the residuals to check for forecast bias
residual_mean_naive_drift <- mean(
  augment(naive_drift_employment_fit)$.resid,
  na.rm = TRUE)
residual_mean_naive_drift
```

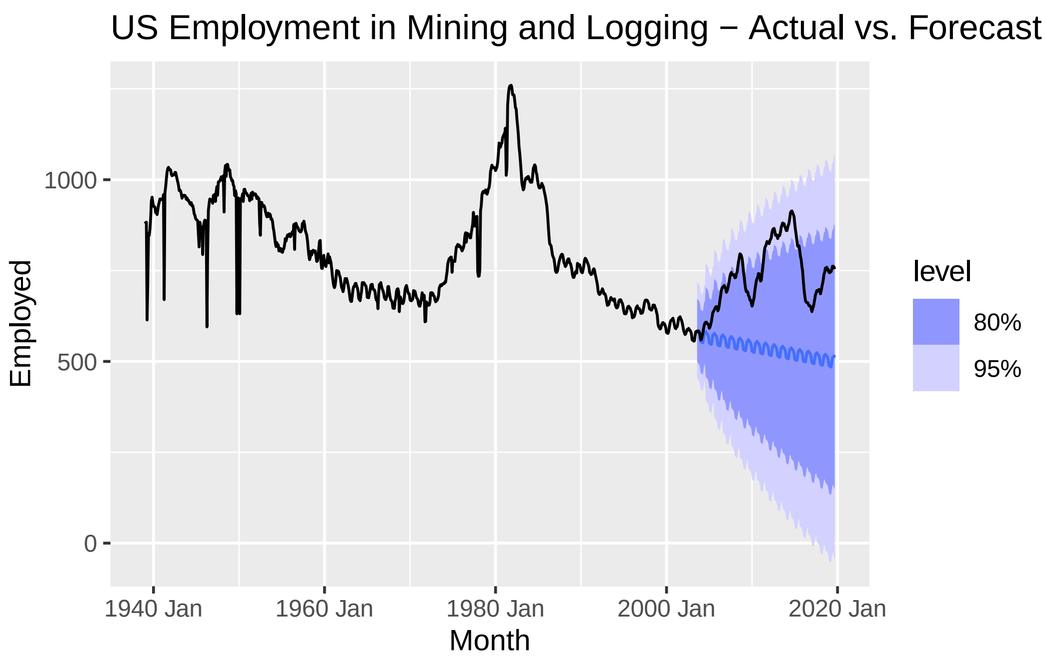
```
[1] -4.170354e-14
```

3.3.4.2 Answering questions to 2nd Benchmark Model

Does the result support your conclusions from (c)?

3.4 Step 4: Evaluate the Point Forecast Accuracy of your 2nd Benchmark Model

```
# Generate forecasts for the test set period
fc_naive_drift <- naive_drift_employment_fit %>%
  forecast(h=(overall_count - training_row_count))

# Plot the forecast values with prediction intervals against the actual complete dataset
fc_naive_drift %>%
  autoplot(mySeries) +
  labs(title="US Employment in Mining and Logging - Actual vs. Forecast (Seasonal Naive wi
```




```
# Calculate point forecast accuracy metrics (MAE, RMSE, MAPE, MASE) on the test set
fc_naive_drift %>% accuracy(mySeries)
```

```
# A tibble: 1 x 11
  .model       Series_ID   .type    ME  RMSE   MAE   MPE  MAPE  MASE RMSSE  ACF1
  <chr>        <chr>       <chr> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 SNaive_Drift CEU1000000~ Test   199.  223.  199.  25.8  25.9  4.56  3.27 0.986
```

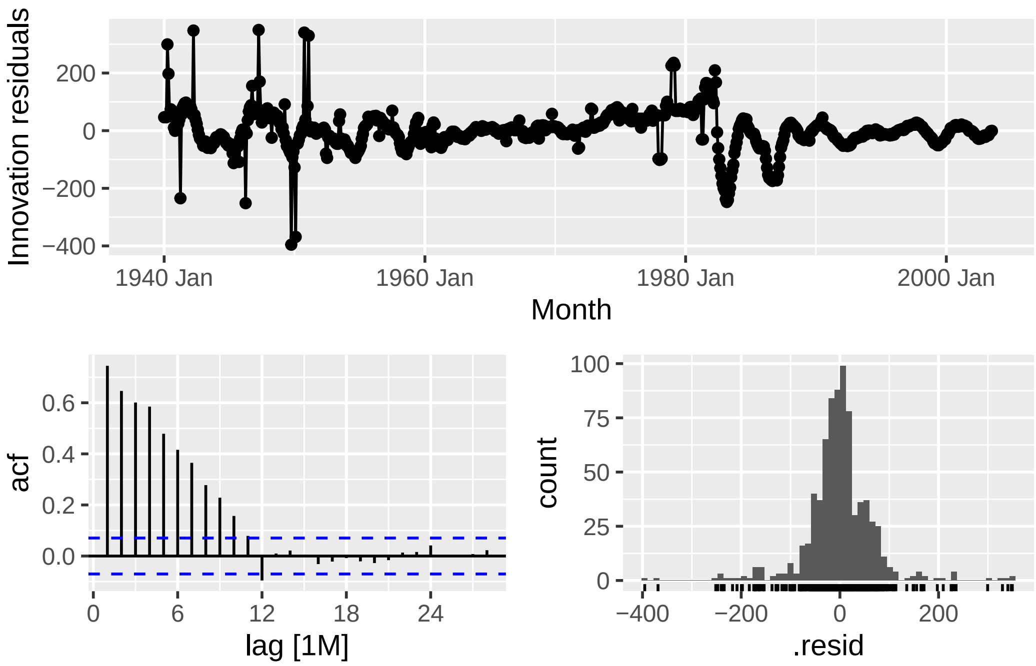
3.4.0.1 Answering questions to 2nd Benchmark Model

Interpret the **RMSE** (Root Mean Squared Error) in max. 3 sentences:

Interpret the **MAPE** (Mean Absolute Percentage Error) in max. 3 sentences:

3.5 Step 5: Evaluate the Point Forecast Uncertainty of Your 2nd Benchmark Model

```
# Perform residual diagnostics to assess homoscedasticity and normality
naive_drift_employment_fit %>% gg_tsresiduals()
```




```
# Extract the upper and lower boundaries for 80% and 95% prediction intervals
fc_naive_drift %>% hilo()
```

```
# A tsibble: 194 x 7 [1M]
# Key:       Series_ID, .model [1]
   Series_ID     .model          Month
   <chr>         <chr>           <mth>
 1 CEU1000000001 SNaive_Drift 2003 Aug
 2 CEU1000000001 SNaive_Drift 2003 Sep
 3 CEU1000000001 SNaive_Drift 2003 Okt
 4 CEU1000000001 SNaive_Drift 2003 Nov
 5 CEU1000000001 SNaive_Drift 2003 Dez
 6 CEU1000000001 SNaive_Drift 2004 Jan
 7 CEU1000000001 SNaive_Drift 2004 Feb
 8 CEU1000000001 SNaive_Drift 2004 Mär
 9 CEU1000000001 SNaive_Drift 2004 Apr
10 CEU1000000001 SNaive_Drift 2004 Mai
# i 184 more rows
# i 4 more variables: Employed <dist>, .mean <dbl>, `80%` <hilo>, `95%` <hilo>
```

3.5.0.1 Answering questions to 2nd Benchmark Model

Are the residuals homoscedastic? (That is, do they have constant variance?)

Are the residuals normally distributed?

Does it make sense to include these prediction intervals in your model evaluation? Why/why not?

4 Compare the Evaluation Results of the Two Models